| Problem Chosen | 2026 MCM/ICM Summary Sheet | Team Control Number |
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# Your Paper's Title(SHUMOJIAYOUZHAN)


**Summary**

hellow everyone



**Keywords**: keyword1; keyword2

# Contents

# 1 Introduction

In professional sports, a long-standing tension exists between the pursuit of athletic glory and the necessity of financial sustainability. While fans focus on the "tip of the iceberg"—the players on the court—owners must manage the complex business machinery beneath. For leagues like the Women's National Basketball Association (WNBA), this challenge is magnified as they transition from risky startups to major entertainment enterprises.

Our objective is to design a dynamic decision-making model that helps owners balance these competing goals. We focus on the following two core dimensions of value:

- **On-Court Performance Index ($OPI$):** Measuring a player's direct contribution to winning games through advanced statistics.
- **Commercial Value Index ($CVI$):** Quantifying a player's ability to generate revenue via ticket sales, jersey sales, and media engagement.

The complexity of modern sports management stems from several volatile factors:

- The non-linear relationship between winning games and generating profit.
- The temporal element of player value, including future potential and injury risks.
- Market dynamics influenced by team location and league-wide constraints like salary caps.

## 1.1 Our Approach

In this paper, we propose a multi-objective optimization framework to guide team management. By integrating $OPI$ and $CVI$, we provide a roadmap for roster construction and financial leveraging.

**Theorem 1.1.** *Optimal Roster Efficiency occurs when the marginal cost of $OPI$ equals the marginal revenue generated by $CVI$, subject to the league salary cap.*

## 1.2 Core Assumptions

To ensure the model is both robust and applicable, we establish the following assumptions:

- **Rational Management:** Owners aim to maximize the combined utility of team profit and asset value[cite: 44].
- **Data Availability:** Publicly available sports and financial data are treated as accurate representations of past performance[cite: 36].

- **Systemic Constraints:** League rules, such as salary caps and roster limits, remain constant unless an expansion is explicitly modeled[cite: 51].
- **Market Liquidity:** Players can be traded or acquired through standard practices like free agency or drafts[cite: 46, 91].

# 2 Assumptions and Justifications

# 3 Notations

# 4 The Dual-Index Model

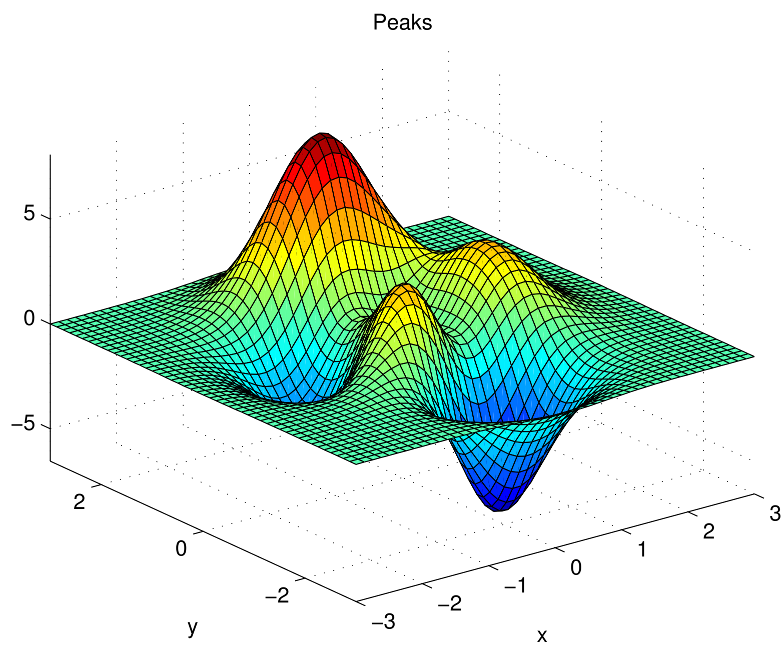


Figure 1: aa

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$$p_j = \begin{cases} 0, & \text{if } j \text{ is odd} \\ r!\,(-1)^{j/2}, & \text{if } j \text{ is even} \end{cases}$$

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$$\arcsin\theta = \oiiint_{\varphi} \lim_{x\to\infty} \frac{n!}{r!\,(n-r)!} \tag{1}$$

# 5 Task-Specific Models

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# 6 Sensitivity Analysis

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# 7 Model Evaluation and Further Discussion

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## 7.1 Strengths

- **Applies widely**
  This system can be used for many types of airplanes, and it also solves the interference during the procedure of the boarding airplane,as described above we can get to the optimization boarding time.We also know that all the service is automate.

- **Improve the quality of the airport service**
  Balancing the cost of the cost and the benefit, it will bring in more convenient for airport and passengers.It also saves many human resources for the airline.

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# 8 Conclusion

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# References

[1] D. E. KNUTH The TEXbook the American Mathematical Society and Addison-Wesley Publishing Company , 1984-1986.

[2] Lamport, Leslie, LATEX: " A Document Preparation System ", Addison-Wesley Publishing Company, 1986.

[3] http://www.latexstudio.net/

[4] http://www.chinatex.org/

# Appendices

## Appendix A First appendix

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl. Here are simulation programmes we used in our model as follow.

**Input matlab source:**

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
    if pab(i)<0.4
       aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

## Appendix B Second appendix

some more text **Input C++ source:**

```
//===========================================================================
// Name        : Sudoku.cpp
// Author      : wzlf11
// Version     : a.0
// Copyright   : Your copyright notice
// Description : Sudoku in C++.
//===========================================================================

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
```

```
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
        for(int y = 0; y < 9; y++){
            cout << table[x][y] << " ";
        }

        cout << endl;
    }

    return 0;
}
```